SYSTEM SPECIFICATION DOCUMENT

Import Data Using Transform Maps

Detailed technical requirements for importing spreadsheet data, mapping staging fields, validating data integrity, and automating target database synchronization.

CORE REQUIREMENTS OVERVIEW

1. File Ingestion & Parsing

- Support XLSX, XLS, and CSV spreadsheet formats.
- Automate staging table generation from uploaded header rows.

2. Role-Based Data Import

- Assign import privileges based on administrative role.
- Restrict unauthorized upload of bulk database records.

3. Transform Map Management

- Create, update, and remove field mapping configurations.
- Define source-to-target field relationships effortlessly.

4. Mapping Permissions & Logic

- **Read:** Preview staging data before transformation.
- **Write:** Modify mapping fields & coalesce rules.
- **Execute:** Run transform scripts and insert/update.

DATA VALIDATION & PIPELINE CONTROLS

5. Invalid Row & Error Handling

- Automatically reject rows failing schema rules.
- Prevent corrupt spreadsheet records from breaking target tables.

6. User & Import Management

- Administrators manage user import privileges.
- Assign roles for staging table access and execution.

7. Transformation Activity Monitoring

- Log every transform map run, row count, and error count.
- Provide searchable logs for audit and compliance teams.

8. Custom Mapping Requests

- Users can request custom transform scripts for complex data.
- Administrators review and approve transformation rules.

DATA SECURITY & EXECUTION ARCHITECTURE

9. Security & Coalesce Protection

- Protect sensitive target fields during bulk spreadsheet ingestion.
- Enforce mandatory coalesce fields (unique identifiers) to prevent accidental duplicate insertions.
- Ensure consistent validation, data scrubbing, and security policies throughout the entire transform pipeline.

10. ADMINISTRATIVE DASHBOARD CAPABILITIES

Module	Functionality	Access Level
Spreadsheet Staging	Upload file, inspect raw staging records, detect data types.	Import Administrator / Data Analyst
Transform Maps	Auto-map matching column names, set field-level transform scripts.	System Administrator
Coalesce & Deduplication	Specify primary key fields to match existing records for updates.	System Administrator
Execution Logs	Review execution duration, success counts, and error stack traces.	Audit & Security Administrators

SYSTEM ARCHITECTURE WORKFLOW

Data Flow Sequence:

1. **Spreadsheet Upload:** User submits .xlsx / .csv file to the Staging Table.
2. **Transform Map Definition:** Field-to-field mapping applied with optional transformation scripts.
3. **Coalesce Evaluation:** Target table searched for matching unique key records.
4. **Record Update / Insertion:** Existing records updated; new records inserted safely.
5. **Audit Logging:** Execution log generated with complete row-level status details.

Executive Conclusion

The solution will provide a centralized, automated Transform Map mechanism for spreadsheet imports, ensuring data from external spreadsheets is cleanly mapped, validated, deduplicated, and accessible only to authorized users according to strict role-based controls.